

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : C-14.

Team Members :

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1. Problem Title : Inventory Management System for Retail Stores

2. Problem Description : To solve this, an **Inventory Management System** will be developed using **Java**. The system will allow users to **add, update, delete, and search products**, while keeping track of **product ID, name, price, and quantity in stock**. It will also **automatically update stock after sales** and alert the user when stock levels are low.

3. Identified Entities (Classes)

Class Name	Description
Product Class	Stores product details (ID, name, price, quantity).
Supplier Class	Stores supplier information. store inventory.
Inventory Class	Manages products and stock operations.
Main Class	Runs and demonstrates the program

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Product class	Int id, String name, double price, int quantity	+ Product() + displayProduct () :void
Supplier Class	int supplierId, String supplierName ,	+ Supplier() + displaySupplier() : void
Inventory Class	ArrayList<Product>	+addProduct():void +displayInventory(): void +sellProduct():void +restockProduct(): void
Main Class	None	Main()

